K S VENKATRAM

Phone: +91 6380150516 | Email: venkatram.ks@gmail.com | Location: Chennai, Tamil Nadu, India
GitHub: github.com/venkatramks | Organization: github.com/Vetri-Namathey | LinkedIn: linkedin.com/in/venkatram-krishnapuram

Education

- **Amrita Vishwa Vidyapeetham** *2023 – 2027*
  B.Tech in Computer Science and Engineering - Artificial Intelligence — CGPA: 7.99/10 Coimbatore, India
- **Class XII – 89.6% | Class X – 87.8%** *Chennai, India*

Work Experience

- **Computer Vision Engineer Intern** *Apr 2026 – Jun 2026*
  Cloverbridge Technologies Pvt Ltd Chennai, India
  - Built and benchmarked a 10-stage Person Re-Identification pipeline spanning detection, tracking, embedding, and gallery-matching (YOLOv8, RT-DETR, StrongSORT, OSNet, TransReID, InsightFace).
  - Designed 5 independent anti-swap triggers (bounding-box IoU, occlusion overlap, match-confidence margin) that reduced identity switches by **70%** in multi-camera, high-occlusion scenes.
  - Trained a lightweight GRU embedding refiner with triplet loss and hard-negative mining, widening the intra/inter-person similarity gap from 0.12 to 0.26; deployed modular CV pipelines with FastAPI and React.
- **AI Engineer Intern** *Apr 2025 – Jul 2025*
  DigiSignals India Pvt Ltd Chennai, India
  - Contributed to an Agentic RAG application for Bray International, enabling AI-based question answering over enterprise documents.
  - Implemented a feedback analysis workflow to flag inaccurate AI-generated responses and improve response reliability.

Projects

- **NeuroVision AI - 3D Brain Tumor Segmentation** *Feb 2026*
  Python, PyTorch, 3D U-Net, FastAPI, React, TypeScript, Tailwind CSS GitHub
  - Built a 3D medical imaging system for brain tumor segmentation using multi-modal MRI volumes and a 3D U-Net model.
  - Added Dice/IoU-based evaluation, uncertainty estimation, backend inference APIs, and React-based result visualization.
- **Hand Gesture-Driven Speech Aid for Mute Individuals** *Published, IEEE*
  Python, Arduino, KNN, PySerial Paper
  - Co-authored and published an IEEE paper on a real-time gesture-to-speech system, using flex sensors and an accelerometer on Arduino Mega to capture hand movement as raw sensor data.
  - Built a K-Nearest Neighbors classifier (Minkowski distance) in Python to classify gestures from sensor input, achieving **90.91% accuracy** on the test set.
  - Implemented a PySerial pipeline for real-time communication between the Arduino hardware and the Python classification/speech-output backend.

Skills

- **Languages:** Python, C++, C, SQL, JavaScript, TypeScript
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, Large Language Models, RAG, Prompt Engineering
- **Libraries/Frameworks:** PyTorch, OpenCV, NumPy, Pandas, scikit-learn, FastAPI, Flask, React, Next.js, Tailwind CSS
- **Databases/Tools:** PostgreSQL, MySQL, SQLAlchemy, Docker, Git, GitHub, Postman, Azure App Service, Azure Logic Apps

Certificates

- **Machine Learning and Deep Learning – IIT Madras, June 2024:** View Certificate
- **Large Language Models – IIT Madras, June 2024:** View Certificate
- **Computer Vision Internship – Cloverbridge Technologies, Jun 2026:** View Certificate